Linux Basics Assignment Documentation

Student Name: Ansari Zakaullah Muhammed Muslim

Task 1: Creating and Renaming Files/Directories

- **Objective:** Practice creating directories, empty files, and renaming files using basic navigation commands.

```
WIN-Q392S61OPE5:~$ mkdir test_dir
WIN-Q392S61OPE5:~$ cd test_dir/
WIN-Q392S61OPE5:~/test_dir$ ls
WIN-Q392S61OPE5:~/test_dir$ ls -a
.  ..
WIN-Q392S61OPE5:~/test_dir$ touch example.txt
WIN-Q392S61OPE5:~/test_dir$ ls
example.txt
WIN-Q392S61OPE5:~/test_dir$ mv example.txt renamed_example.txt
WIN-Q392S61OPE5:~/test_dir$ ls
renamed_example.txt
WIN-Q392S61OPE5:~/test_dir$ ls -l
total 0
-rw-r--r--  1 zaka  zaka  0 Sep  3 09:11 renamed_example.txt
WIN-Q392S61OPE5:~/test_dir$
```

- **Explanation:**
  1. mkdir creates a new directory.
  2. touch creates an empty file.
  3. mv renames the file from example.txt to renamed_example.txt.
  4. ls -l lists the directory contents in long format to verify the change.

Task 2: Viewing File Contents

- **Objective:** Explore system files and view specific portions using file inspection utilities.

```
WIN-Q392S61OPE5:~/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/sh
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/mail:/sbin/nologin
news:x:9:13:news:/usr/lib/news:/sbin/nologin
uucp:x:10:14:uucp:/var/spool/uucppublic:/sbin/nologin
cron:x:16:16:cron:/var/spool/cron:/sbin/nologin
ftp:x:21:21::/var/lib/ftp:/sbin/nologin
sshd:x:22:22:sshd:/dev/null:/sbin/nologin
games:x:35:35:games:/usr/games:/sbin/nologin
ntp:x:123:123:NTP:/var/empty:/sbin/nologin
guest:x:405:100:guest:/dev/null:/sbin/nologin
nobody:x:65534:65534:nobody:/:/sbin/nologin
zaka:x:1000:1000::/home/zaka:/bin/sh
WIN-Q392S61OPE5:~/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/sh
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
WIN-Q392S61OPE5:~/test_dir$ tail -n 5 /etc/passwd
games:x:35:35:games:/usr/games:/sbin/nologin
ntp:x:123:123:NTP:/var/empty:/sbin/nologin
guest:x:405:100:guest:/dev/null:/sbin/nologin
nobody:x:65534:65534:nobody:/:/sbin/nologin
zaka:x:1000:1000::/home/zaka:/bin/sh
WIN-Q392S61OPE5:~/test_dir$
```

- **Explanation:**
  1. cat outputs the entire content of a file.
  2. head -n 5 extracts only the first 5 lines, while tail -n 5 extracts the final 5 lines of the system user database file /etc/passwd.

Task 3: Searching for Patterns

- **Objective:** Search through files for specific patterns or strings using grep.

```
WIN-Q392S61OPE5:~/test_dir$ grep ""root /etc/passwd
root:x:0:0:root:/root:/bin/sh
WIN-Q392S61OPE5:~/test_dir$
```

- **Explanation:**
1. grep scans the target file line-by-line and prints out any lines containing the matching search string "root".

Task 4: Zipping and Unzipping

- **Objective:** Archive directories and extract compressed files.

```
WIN-Q392S61OPE5:~/test_dir$ cd ..
WIN-Q392S61OPE5:~$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
WIN-Q392S61OPE5:~$ ls
test_dir      test_dir.zip
WIN-Q392S61OPE5:~$ unzip test_dir
test_dir.zip  test_dir/
WIN-Q392S61OPE5:~$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
 extracting: unzipped_dir/test_dir/renamed_example.txt
WIN-Q392S61OPE5:~$ ls
test_dir      test_dir.zip  unzipped_dir
WIN-Q392S61OPE5:~$ cd unzipped_dir/
WIN-Q392S61OPE5:~/unzipped_dir$ ls
test_dir
WIN-Q392S61OPE5:~/unzipped_dir$ cd ..
WIN-Q392S61OPE5:~$ ls -la unzipped_dir/
total 3
drwxr-sr-x    3 zaka     zaka          1024 Sep  3 09:26 .
drwxr-sr-x    4 zaka     zaka          1024 Sep  3 09:26 ..
drwxr-sr-x    2 zaka     zaka          1024 Sep  3 09:12 test_dir
WIN-Q392S61OPE5:~$
```

- **Explanation:**
  1. zip -r recursively compresses a directory into a .zip archive.
  2. unzip -d extracts the archive contents into a newly specified target directory, which is then verified with ls -la.

Task 5: Downloading Files

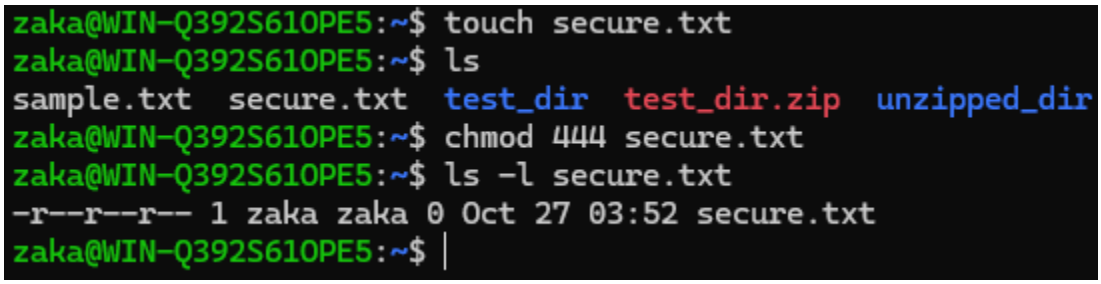
- **Objective:** Download files directly from the web using command-line networking tools.



- **Explanation:**
  1. wget fetches files from web URLs over HTTP/HTTPS.
  2. The -O flag allows renaming the downloaded output file locally to sample.txt.

Task 6: Changing Permissions

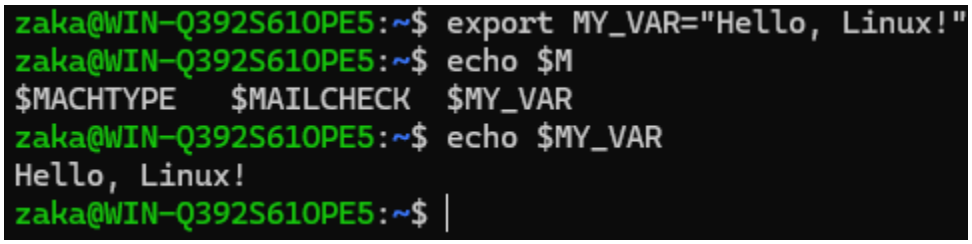
- **Objective:** Modify Linux file permission bits to restrict or grant access.



- **Explanation:**
  1. chmod 444 sets numeric permissions where 4 grants read-only permission (r--) to the owner, group, and others respectively, protecting the file from modifications.

Task 7: Working with Environment Variables

- **Objective:** Define and reference shell session environment variables.



- **Explanation:**
  1. export sets a new session-level environment variable.
  2. echo retrieves and prints the stored value using the $ symbol prefix.

GitHub Link: https://github.com/zakaullah-ansari/Linux_Basics_Assignment.git